



Laypersons' understanding of statistical concepts commonly used in prescription drug promotion: A review of the research literature

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ABSTRACT

Background: The prevalence of direct-to-consumer (DTC) advertising for prescription drugs has led to concerns about how consumers interpret the medical information conveyed in these ads. One strategy for improving lay understanding of medical information involves incorporating quantitative information about a treatment's potential benefits and risks.

Objective: This literature review investigates laypersons' interpretations of statistical concepts, expanding on past reviews and including terms that may be used in DTC prescription drug advertising.

Methods: We searched six databases for articles published from January 2000 to October 2021. Articles were included if they were in English and examined general or lay audiences' comprehension of quantitative or statistical concepts, without limiting the context of the studies to medical situations.

Results: We identified 25 eligible articles. The evidence suggests that likelihood ratios, odds ratios, probabilities, numbers needed to treat/harm, and confidence intervals hinder comprehension of quantitative information. The results are mixed for information presented as frequencies, percentages, absolute risk reduction, and relative risk reduction. The mixed findings could be due to numeracy, framing as risks or benefits, and operationalization of the outcomes. We found no studies examining interpretations of minimum, maximum, central tendency, power, statistical significance, or hazard ratio.

Conclusion: Studies spanning several decades have examined how laypeople interpret statistical concepts. While a few terms are consistently studied, many questions still remain on how to make risk information more understandable to lay audiences, particularly those with low numeracy.

1. Introduction

Over the past 20 years, there has been a rapid increase in direct-to-consumer (DTC) prescription drug advertisements.^{1,2} This surge has also given rise to increasing concern about whether DTC promotional materials misinform consumers by overemphasizing benefits and underemphasizing risks.^{2,3} Equipped with better understanding of medical information, laypeople are able to make better informed decisions. Emerging research has focused on identifying best practices for communicating medical treatment information to laypeople.^{4–7}

One strategy for improving the lay understanding of medical

information involves incorporating quantitative information about a treatment's potential benefits and risks.^{4,5} Studies have found that adding accurate quantitative information to DTC promotion, such as presenting information in structured formats like percentages, helps laypeople understand how well a drug works.^{6,7} Indeed, some DTC promotion already includes quantitative information; for instance, one study found that 60 % of oncology DTC websites provided quantitative benefit information and 49 % provided quantitative risk information.⁸

Previous literature reviews have identified best practices for communicating health risks but often focus on specific statistical concepts or rely on expert opinions. For example, Büchter et al.'s⁹ review found that presenting the probability of adverse effects to laypeople

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List of abbreviations	
ASSIA	Applied Social Science Index and Abstracts
ARR	Absolute risk reduction
DTC	Direct-to-consumer
FDA	U.S. Food and Drug Administration
NNH	Number needed to harm
NNT	Number needed to treat
RRR	Relative risk reduction

numerically can result in more accurate estimations of risk compared with using qualitative descriptions (e.g., “common” or “uncommon”). In another review, Akl et al.¹⁰ focused on the impact of using different formats to present risk information (e.g., frequencies, percentages, or probabilities) and risk reductions (e.g., relative risk reduction [RRR], absolute risk reduction [ARR], or number needed to treat [NNT]) on understanding diagnostic and screening tests among laypeople and health care professionals. The authors found that frequencies (e.g., “50 in 1000”) are better understood than percentages. They also found that RRR (e.g., “Cuts the risk of fracture by 50 %”) is more persuasive than ARR (e.g., “5 % less would suffer a fracture”) and NNT but that the RRR may lead to misinterpretation. Other reviews have found similar issues with presenting RRR. Visschers et al.¹¹ cautioned against presenting RRR, noting that it can lead to confusion and may be mistaken for ARR. Another review found that absolute risk formats and visual aids can improve laypeople’s understanding of probabilistic information.¹²

Although some evidence of best practices exists, researchers have also identified areas in need of further exploration. For example, in a literature review, Han¹³ concluded that more evidence is needed to determine specific recommendations on how best to represent and communicate uncertainty in clinical contexts. More recently and as an update to a previous literature review, Bonner et al.¹⁴ gave recommendations for presenting information about risks and benefits in patient decision aids such as using consistent denominators, using consistent risk formats (e.g., percentages, frequencies), and considering readers’ needs with varying numeracy. However, the researchers noted that best practices for communicating other topics—like conveying uncertainty or risk over time—are still to be determined.

In 2023, the U.S. Food and Drug Administration (FDA) released final guidance providing recommendations for presenting quantitative efficacy and risk information in DTC promotional materials.¹⁵ The guidance recommends using frequencies (e.g., “55 out of 100”) or percentages (“55 %”) instead of relative frequencies (e.g., “33 % reduction in symptoms”). If using relative frequencies, the guidance recommends using them in conjunction with absolute frequencies to maximize consumer understanding. The guidance also provides recommendations on formatting this quantitative information, including using a consistent format for numerical information, with whole numbers and denominators in multiples of 10, if possible. In addition, visual aids—such as graphs, tables, and icon arrays—may be used to improve consumers’ comprehension. While the guidance provides some recommendations on presenting efficacy and risk information, not all probability formats were included in the guidance. More research is needed to determine how consumers interpret risk information presented with different formats such as medians, confidence intervals, ranges, and odds ratios.

While it is generally understood that laypeople are more likely to understand risk and benefit information when it is presented numerically than when it is presented non-numerically,⁵ how they interpret specific quantitative claims is unclear. Further questions remain about the best ways of conveying numeric information to laypeople to help them make healthcare decisions. Leveraging the findings from previous studies, this literature review aims to describe how laypeople interpret a variety of quantitative terms and phrases that could potentially be used

in DTC promotional materials. This review aims to broaden the array of statistical concepts examined by addressing confidence intervals, likelihood ratios, and central tendencies (e.g., averages, means and medians). We also examine the interpretation of these concepts through a variety of scenarios, from marketing to social interactions, without limiting the scenarios to medical situations in order to broaden the search. In addition, this review identifies potential gaps in the current research literature on how laypeople understand statistical concepts.

2. Methods

2.1. Data sources

A comprehensive search was conducted between September and October 2021 in the following databases: PubMed, APA PsycInfo®, ScienceDirect, Applied Social Science Index and Abstracts (ASSIA), Social Science Premium Collection (Education Collection and Social Science Database), and Business Source Premier. Our search strategy was constructed using terms from primary subject domains including, but not limited to, statistics/mathematics, quantitative literacy, statistical literacy, comprehension, and risk communication. Searches were tailored to the syntax of each database using Boolean operators, and keywords were truncated to retrieve variations of those terms. A full listing of the search terms is available in [Supplemental Table 1](#).

2.2. Inclusion and exclusion criteria

The inclusion criteria incorporated peer-reviewed literature and full-text articles on empirical studies relevant to statistical literacy or the understanding or perceptions of quantitative terms and phrases published from January 2000 to October 2021. The articles had to be in English and focus on general adult or adult lay audiences. We excluded review articles, publications in a non-English language, editorials, commentaries, letters, book reviews, conference abstracts with no full text, proprietary documents, recommendations or guidelines without research, literature outside of the scope of the research questions, studies that primarily focused on the graphical display of quantitative information (e.g., graphs, charts, and figures), and publications that focused on medical or research professionals, educators, or youth.

2.3. Study selection

Abstracts and studies were independently reviewed by two research team members using the criteria mentioned above. Discrepancies between reviewers were resolved through discussions and consultation with a third team member. Abstracts without enough information to assess the inclusion/exclusion criteria were included for full-text review. Similarly, a third reviewer reconciled disagreements during the selection of full-text articles.

2.4. Data extraction

The researchers independently reviewed the final articles and extracted data for the following variables: first author, year of publication, country of study, study context(s), statistical/quantitative term(s), respondents, study design(s), independent variables, primary outcome measure(s), and main finding(s). The data extracted was reviewed by a second team member, and the two authors met to discuss and reconcile any differences from the data.¹⁶

2.5. Data synthesis

We organized our findings based on the statistical concepts found in the literature to summarize how laypeople comprehend or view different statistics. We grouped the statistical concepts based on types of probability statements: those providing benefit or risk information in

general (e.g., frequency, percentages, or single or multiple event probabilities) and those comparing one treatment or drug to another treatment, drug, or control group (e.g., likelihood ratio, ARR, or odds ratio) (see Table 1). Further, studies often examine similar types of statistics together; therefore, our results are presented with terms often studied together (see Table 2). We also present descriptive characteristics and the articles' main findings in Table 3. This literature review applied principles of the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA), to document our searching, study selection, and reporting of findings. Our review protocol was registered with the open source platform OSF (<https://osf.io/brks6>).¹⁷

3. Results

3.1. Article selection

We identified 1384 articles, of which 303 were duplicates and therefore removed. After reviewing the titles and abstracts, 947 were removed due to lack of relevance (e.g., they mentioned a statistical term, but it only relayed the results of a study). The remaining 134 articles underwent a full-text review for eligibility. During this review, we made a post hoc change to the eligibility criteria discussed in the protocol. In order to not duplicate previous efforts and instead focus on novel findings, we excluded 10 articles published before 2011 that had been included in previous literature reviews.^{18–27} We do not believe that excluding these articles would result in severe bias as they are summarized in previous reviews discussed here.^{5,9–14} Furthermore, we retained two pre-2011 articles^{28,29} that were not included in the previous review articles. The final inclusion and exclusion criteria resulted in 25 articles for review.^{28–52} Fig. 1 is a PRISMA diagram outlining how the search of databases and review of articles resulted in the final 25 in-scope articles.

There were nine main statistical concepts discussed in the literature, though they are often referred to in articles using different terminology. For simplicity, Table 1 provides the statistical concepts and notes the related terms used in the literature; describes how the term is often presented or operationalized; and offers a brief description.

As shown in Table 2, the most common type of study in this review examined frequencies in comparison to percentages, and frequencies with variable denominators (e.g., 1 in X), followed by confidence intervals versus point estimates.

Table 3 provides the list of eligible articles included in this literature search, the statistical concept(s) examined, and their main findings.

3.1.1. Frequencies and percentages

The studies comparing frequencies and percentages provided inconsistent findings. Three studies showed no differences between these formats on the perceived risk of the side effects of taking a medication,⁴² ease of understanding or level of confusion,⁴⁹ or on perceptions of products/services.⁴⁵ Several studies provided support for the advantages of percentage formats.^{31,45,52} Woloshin and Schwartz⁵² found that participants were better able to understand percentages than frequencies (e.g., 20 in 1,000, 2 in 100). In a qualitative study, Damman et al.³¹ found that percentages were recalled more than frequencies and that frequencies may be harder to understand. Finally, Sevilla et al.,⁴⁵ who examined the impact of percentage and frequency formats on actual purchasing of cheese, found that more cheese was sold when presented with percentages (e.g., “Out of 2024 entrants, Cheese X is one of 20 % of cheeses to receive an award”) than with frequencies (e.g., “Out of 2024 entrants, Cheese X is one of 411 cheeses to receive an award”). In contrast, two studies showed that frequency formats were clearer, processed faster, and resulted in higher intentions to take action than percentages. Brase²⁸ found that frequencies were perceived as clearer than percentages. Oudhoff and Timmerman⁴¹ found that when participants viewed lottery information as a frequency (e.g., 1 in 10), they processed the information faster, perceived they had a better chance of winning the lottery and were more likely to indicate they would buy the lottery

Table 1
Summary of statistical concepts.

Statistical Concept	Related Term (s) Used in the Literature	Presentation Format(s)	Description
Frequency	Natural	1 out of 10	Presented as a whole number, this is the count of the number of times (or events) something is observed. The count is often accompanied with baseline information describing the total number of times that the event could have occurred, similar to how fractions are presented. A number out of a given number expressed in hundredths; for example 10 out of 1000 translates to 1 percent. The confidence interval provides the range of values, often around a point estimate, where the actual value is expected to lie given a degree of confidence (e.g., 95 %). For example, “The average person’s lifetime chance of developing colon cancer is between 2 % and 10 %.” The point estimate is an estimate of the actual value in the population. For example, “The average person’s lifetime chance of developing colon cancer is 6 %.
	Frequency	5 out of 50	
	Absolute	267 million	
	Frequency Simple	One in a million	
	Frequency 1 in X N in X*N Random match probability (RMP)		
Percentage	Percent	5 %	Single event probability is the likelihood an event will happen (e.g., The probability that a woman has breast cancer is 0.9 %). Conditional probability is the probability of an event happening based on another event or previous events (e.g., if a woman has breast cancer, the probability that she will have a positive mammography test is 90 %).
Confidence Interval and Point Estimate	Range or uncertainty range	6 %: 2%–10 % 225: 200–250	AR is the risk that a person will experience a certain event. ARR is the difference between risk of an event in a comparison group and risk of an event in a treatment group (i.e., risk reduction from 6 % to 3 % is 3 % ARR).
	Best estimate		
Probability	Single Probability Conditional Probability	0.9 %	
Absolute risk reduction (ARR)	Absolute Risk (AR) Absolute Terms	6 % fewer deaths from Treatment A versus a control group Treatment A reduced risk by 10 per 1000. Treatment B reduced risk by 4 per 1000	

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Table 1 (continued)

Statistical Concept	Related Term (s) Used in the Literature	Presentation Format(s)	Description
Relative risk reduction (RRR)	Relative Risk (RR) Relative Terms	Risk reduced by 40 % if patients receive Treatment A Relative Risk: 2 times as many patients taking Drug A will experience risk B	ARR = Control group rate – treatment group rate. In healthcare, the RR is typically defined as the risk of a certain outcome under treatment compared to the risk of the outcome under the control condition. It is a ratio of two probabilities or risks. RRR is the relative decrease in the risk of an outcome in the treatment condition compared to a control condition (i. e., risk reduction from 6 % to 3 % is 50 % RRR) RRR = (Control group rate – treatment group rate)/control group rate.
Number needed to treat/harm (NNT/NNH)		Treatment A requires treating 100 people to prevent disease in 1 person 10 patients would have to take Drug A for 1 patient to experience Risk B	The number of patients that need to be given a particular treatment to get the desired effect in one individual (e.g., for every 500 patients who take a medication compared to those who do not, one additional patient will show improvement). Note that this is just a different way of providing the ARR as it is 1/ARR. (1/0.03 = 33.33 or 34 patients need to be treated to prevent one event).
Likelihood ratio (LR)		10 times more likely	The ratio of two likelihoods a) the probability that the person with the condition has a certain test result; and b) the probability that the person without the condition has a certain test result. For example: The probability that DNA comes from the same source as a criminal suspect, versus the probability that the DNA does not come from that source (e.g., DNA evidence is 100 times more likely to have come from the suspect than from a randomly chosen person).
Odds ratio (OR)		Odds of risk is X if a person receives treatment B versus A	OR is the odds of the outcomes or event given exposure to a treatment compared to the odds of the outcome or event without the exposure

Table 1 (continued)

Statistical Concept	Related Term (s) Used in the Literature	Presentation Format(s)	Description
			to the treatment (e.g., odds of dying doubles if you receive Treatment B compared to Treatment A).

Caption: This table displays the statistical concept included, the related terms used in the literature, presentation format, and a short description.

Table 2

Statistical concept and number of articles reviewed.

Statistical Concept	Number of Articles (Total = 25)
Probability statements framed as:	
Frequency and percentages	7
Frequency with variable denominators	7
Confidence intervals and point estimates	5
Single event probability	4
Comparative probability statements framed as:	
Absolute risk reduction (ARR)	3
Relative risk reduction (RRR)	
Numbers needed to treat/harm (NNT/NNH)	
Likelihood ratio (LR)	2
Odds ratios (OR)	1

ticket than when that information was presented as a percentage.

3.1.2. Frequencies with varying denominators

Seven articles examined frequencies with varying denominators (e. g., 1 in 10 and 10 in 100).^{29,41,45–48,52} A common finding was that varying denominators led to lower comprehension than when using consistent denominators.⁵² When using consistent denominators, small denominators led participants to have a higher perceived probability of getting a disease or side effect and reporting intentions not to act, such as to not use a medication or take a trip.^{47,48} Similarly, formats using small denominators (e.g., 1/10) led to higher perceived risk²⁹ compared to formats using large denominators (e.g., 10/100); however when the numerator was very small, the effect was the opposite. That is, participants perceived a higher risk when the numerator and denominator were large (i.e., 90.3/100,000) compared with a very small numerator (i.e., 0.903/1000).²⁹ Only one article found no differences in comprehension due to varying denominators.⁴⁶ When framed as benefits (e.g., likelihood of winning the lottery) rather than risks, a smaller denominator had the expected impact of participants perceiving a higher chance of winning the lottery and being more willing to buy a lottery ticket.⁴¹ Relatedly, Sevilla et al.⁴⁵ found that when shown rankings of products (e.g., top 20 out of 50 or top 200 out of 500), participants perceived the smaller number as better performing even though the percentages from those numbers are equivalent. In addition, they found that participants had more favorable impressions about *frequencies* when the number of possible products or services (i.e., set sizes) were smaller (e.g., out of 50) but more favorable impressions about *percentages* when set sizes were larger (e.g., out of 500) even though they were, again, equivalent percentages.

3.1.3. Confidence intervals and point estimates

Five articles examined the interpretation of confidence intervals and point estimates.^{33,34,38,39,50} These studies fell into two general types: how participants interpreted confidence intervals and how participants interpreted point estimates and confidence intervals.

Two articles examined how participants interpreted confidence intervals. Dieckmann et al.³³ found that in one study, participants interpreted values in the range/confidence interval as equally likely

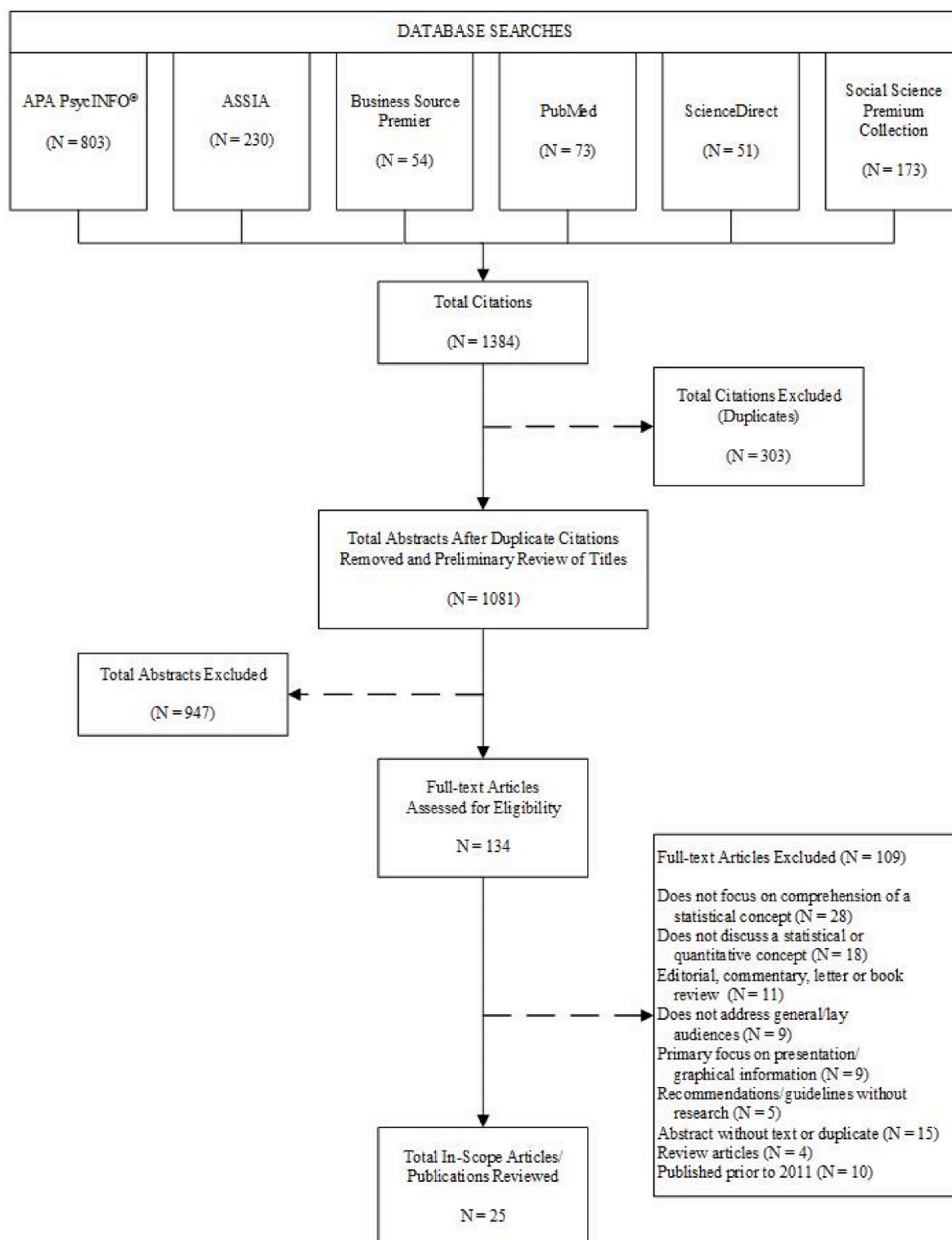


Fig. 1. PRISMA flow diagram for literature search of understanding quantitative concepts.

(uniform) with fewer perceiving values closer to the point estimate as more likely (normal distribution) and even fewer thinking values at the interval ends as more likely. In contrast, in their second study, most participants perceived the distribution as normal. The authors found that participants were more likely to perceive a uniform distribution when no point estimate was provided and when there was no mention of the confidence level (e.g., 90 % confident). Løhre and Teigen³⁸ examined whether participants understood that wider intervals were

associated with higher confidence levels (i.e., 90 % or 60 % confidence level). Results were mixed, with some participants estimating the width of the confidence interval to be larger with a higher confidence level and others estimating it to be larger with a smaller confidence level.

Three studies compared interpretations of confidence intervals and point estimates. Two studies found that participants answered more questions correctly when presented with point estimates than with confidence intervals.^{39,50} The results were similar when the estimates

Table 3

Article characteristics and summary findings.

First Author (Year) Country	Study Context	Statistical Concept (s)	Respondents (N)	Design	Independent Variables	Primary Outcome Measures	Main Findings
Bali, Martire, and Edmond (2021) ³⁰ United States Only Study 2 is included here.	Hypothetical sexual assault case with DNA evidence	- Frequency (random match probability (RMP)) (e.g., one in a million) - Likelihood ratio (LR) (e.g., 5500 times more likely)	Laypeople – web panel Study 2 (204)	Between-subjects (format)	- Format (RMP, probability, LR) - Numeracy	Comprehension - Consistency, Ability, Sensitivity, Orthodox, and Coherence (CASOC) Scores	- No differences in total CASOC score by format. - For coherence, fewer participants correctly identified both fallacy statements in the LR than RMP conditions. - Numeracy was positively related with the total CASOC score.
Brase (2002) ²⁸ United Kingdom	Disease prevalence, education, marketing, drug efficacy, risk of bacteria in meat scenarios, wage discrimination	- Simple frequencies (e.g., 1 in 3) - Relative frequencies/percentages (e.g., 33 %) - Absolute frequencies (e.g., 2.7 million or 267 million) - Single event probabilities (e.g., 0.33)	Undergraduate students Experiment 1 (120) Experiment 2 (122) Experiment 3 (86) Experiment 4 (73)	Experiment 1: Between-subjects Experiment 2: Within-subjects and between-subjects Experiments 3 & 4: All received same scenario, counter-balanced	All experiments - Number format - Magnitude Experiment 2 added reference class for absolute frequencies.	Experiments 1 & 2: - Perceived clarity - Perceived impact (e.g., seriousness) Hypothetical money to allocate Experiment 3: - Rated most serious or least serious Experiment 4: - Who is trying to exaggerate or minimize situation - Perceived clarity	- Simple frequencies consistently rated clearest, followed by percentages. - Probabilities were rated least clear, least serious, and trying to minimize situation. - Absolute frequencies rated most serious, used to exaggerate situation, most impressive when the magnitude was low but less impressive when magnitude was high.
Damman et al. (2017) ³¹ Netherlands	Online cardio-metabolic risk calculator	- Percentage (e.g., 14 %) - Frequency (e.g., 14 of 100) - Comparative risk information (e.g., “Risk for someone like you without risk factors is 10 %”)	Laypeople – age 45–60 (16)	Qualitative study (semi-structured interview)	N/A	- Attention: Used eye tracker - Recall: Completion of a recall task - Perception of risk	- Attention: Gazed most at frequency, which may mean less understanding. - Recall: Percentages recalled better than frequencies. - Interpretation: Participants relied on their knowledge rather than risk information provided.
DeWolf et al. (2014) ³² United States	Magnitude comparison	Magnitude of number type - Fractions - Decimals - Three-digit integers	Students Study 1 (69) Study 2 (95) Study 3 (26)	Studies 1 & 3: Within-subjects (number type) Study 2: Between-subjects (number type)	Study 1: - Decimals, fractions Studies 2 & 3: - Fractions, decimals, integers	- Percent error (if the number was correctly categorized as larger or smaller than comparison) - Response time (RT)	- No effect for order of number type. - Error rates were higher for fractions than for decimals or integers. - RTs were faster for decimals and integers than fractions.
Dieckmann et al. (2015) ³³ United States Only Studies 1 and 2 are included here.	Hypothetical scenarios of wildfires and climate change forecasts	- Range/confidence interval (C.I.) - Confidence level (e.g., 90 % confident) - Point estimate	Laypeople – web panel Study 1 (165) Study 2 (89)	Study 1: Between-subjects Study 2: Within-subjects and between-subjects	Study 1: - Point estimate - Range Study 2: - Point estimate - Range - Confidence level (CL)	Perception of distribution of the C.I. (i.e., all values equally likely (uniform), values in middle more likely (normal), values at end more likely (U-shaped))	- Study 1: Most interpreted distribution as uniform, followed by normal, then U-shaped. - Study 2: Perceived distributions as normal (unlike Study 1). - Participants in the 90 % CL and point estimate condition more likely to perceive as normal, while others more likely to perceive as uniform. - Participants were least likely to perceive U-shaped distribution, but more participants did with no point estimate.
Han et al. (2011) ³⁴ United States Only Experiment 1 is included here.	Hypothetical colorectal cancer risk estimates	- Confidence interval (C.I.) - Point estimate - Comparative risk information	Laypeople – web panel Experiment 1 (240)	Between-subjects (statistical concept) Within-subjects (comparative risk information)	- C.I. - Point estimate - Numeracy	- Perceived cancer risk - Cancer-related worry - Perceived credibility	C.I. related to higher cancer-related worry than point estimates but not perceived risk, even with comparative risk information.
Ilic and Green (2012) ³⁵ Australia	Prostate cancer treatment	Number needed to treat (NNT) to prevent a death	Laypeople – Men (76) Some with	Qualitative study (separate focus groups for	- NNT - OR - RRR	Understanding of information	Strong preference for ARR over other formats.

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Table 3 (continued)

First Author (Year) Country	Study Context	Statistical Concept (s)	Respondents (N)	Design	Independent Variables	Primary Outcome Measures	Main Findings
		• Odds ratio (OR) (e.g., odds of dying doubles) • Relative Risk Reduction (RRR) (e.g., risk of dying is reduced by 40 %) • Absolute Risk Reduction (ARR) (e.g., 5.7 % less deaths)	diagnosed prostate cancer (42) and some without (34)	men with and without prostate cancer)	• ARR	• Preference for framing (surviving/dying)	• Participants reported NNT, OR, and RRR were confusing and did not provide enough information to make a decision. • RRR was more informative than NNT and OR. • Participants preferred positive framing.
Kupor and Laurin (2020) ³⁶ Only 8 out of 10 studies are included here.	Study 1A: Hypothetical drug side effect Studies 1B–1F, Study 2: Vitamins/Minerals in fruit/vegetable resulting in an outcome Study 4A: Advertisement for eyelash-lengthening product	Magnitude (smaller/larger) for: • Frequency • Percentage	Students Study 1A (110) Laypeople – web panel Studies 1B & 1F (202) Studies 1C, 1D & 1E (200) Study 2 (400) Study 4A (204)	Studies 1A–1F & Study 3: Within-subjects Study 2 and 4A: Between-subjects	Study 1A–F, Study 2: • Percentage • Frequency • Magnitude Study 4A: • Magnitude - percentages	Study 1A–1F, Study 2: • Perception of how much participant thought product A led to outcome B Study 4A: • Magnitude perceptions • Behavioral intentions to buy or recommend	• Studies 1A–1F, Study 2, Study 4A: Participants in the larger probability conditions (vs. smaller) predicted larger occurrences and magnitude of outcome. • Study 4A: Participants in the larger probability condition were more likely to intend to buy or recommend the product.
LaVallie et al. (2012) ³⁷ United States	Hypothetical benefits of Treatments A & B	• RRR (e.g., Treatment A reduces risk by 25 %) • ARR (e.g., Treatment A reduces risk by 10 per 1000) • NNT (e.g., Treatment A requires treating 100 people to prevent cancer in 1 person)	Laypeople attending dinner – American Indian and Alaska Native (91)	Survey, no variation in order of risk terms	• RRR • ARR • NNT • Numeracy	• Risk comprehension (three items, total correct)	• RRR format had a significantly higher percentage of correct answers. • An equal number of correct responses were for NNT and ARR. • Higher numeracy was associated with correctly interpreting the risk information.
Løhre and Teigen (2017) ³⁸ Norway	Climate change forecasts	• Confidence intervals (C.I.) • C.I. width • Magnitude of confidence level (60 %/90 % probability it is between X and Y)	Students Experiment 1 (90) Laypeople – web panel Experiment 2 (242) Experiment 3 (394) Experiment 4 (192) Experiment 5 (259)	Experiments 1, 3 & 5: Within-subjects Experiment 2 & Experiment 4: Between-subjects	Experiments 1 & 2: • Probability format ([“There is/I am] X% [probability/certain”]) Experiments 3 & 4: • Magnitudes (60 %/90 %) Experiments 3 & 4: • Narrow and wide C.I. • Numeracy Experiment 5: • Controlled for expertise, numeracy	Experiments 1, 2 & 5: • Prediction of upper and lower limits of C.I. based on magnitude of probability Experiments 3, 4 & 5: • Degree of confidence • Prediction of % probability of outcome	• Inconsistent findings—participants did not consistently think 90 % levels would have intervals that were larger than 60 % levels. • If able to compare different confidence levels (within subjects), then more likely to understand high confidence levels led to wider intervals. • Inconsistent findings for numeracy; numeracy was not always related to response.
Longman et al. (2012) ³⁹ Australia	Hypothetical risk from medication	• Point estimate (20 out of 100) • Small range/C.I. (16–24 out of 100) • Large range (8–32 out of 100)	Students (120)	Between-subjects (source of risk) Within-subjects (point estimate, small range, or large range)	• Uncertainty (small or large range) • Source (clinician, drug company)	• Understanding (number of items correct or partially correct) • Perceived risk • Source credibility	• Point estimates were more correctly understood than large or small ranges. • Large ranges had a higher perceived risk and were less credible than point estimates or small ranges.
Mohan, Chandon, and Riis (2015) ⁴⁰ United States	Fuel efficiency, pages per minute printing, wireless internet access cost, pay per task	• Comparing percentage reduction or loss (cost) or increase (benefit) in new product/service (e.g., 50 % more	Laypeople – online panel Study 1 (606) Study 2 (395) Study 3 (606) Study 4 (150)	Study 1: Between-subjects (outcome, framing, change type) and within-subjects:	Study 1: • Framing: cost/benefit • Change type: cost vs. benefit Studies 2–4:	• Accuracy in understanding the best option to pick	• Participants unable to accurately select the better option between percent change loss versus benefit.

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Table 3 (continued)

First Author (Year) Country	Study Context	Statistical Concept (s)	Respondents (N)	Design	Independent Variables	Primary Outcome Measures	Main Findings
		pages per minute versus 45 % less time to print each page)		(percentage change level) Studies 2–4: Between-subjects	- Standard rate information (15 pages/min) - Numeracy		- Higher numeracy led to fewer errors in option selection. - Providing standard rate information reduced option selection errors.
Oudhoff et al. (2015) ⁴¹ Netherlands	Hypothetical likelihood of winning lottery	- Percent (X%) - Frequencies with variable denominators (X in 100, 1000 or 10,000) - Frequencies standardized with the smallest numerator (1 in X) - Magnitude (e.g., 0.14 %, 3.7 %, 25 %)	Students (mostly medical students) (192)	Between-subjects (risk format) Within-subjects (magnitude)	- Risk formats - Magnitude - Numeracy	- Perceived likelihood of winning lottery - Choosing to buy a ticket - Response Time (RT) (for ability to interpret information)	- Perceived likelihood was higher in the 1 in X format. - When magnitude was small, differences in perceived likelihood due to format were smaller. - RT was longest for X in 100 format. - More likely to buy a ticket in 1 in X condition. - No relationship between formats and numeracy.
Peters, Hart, and Fraenkel (2011) ⁴² United States	Hypothetical side effects of medication	- Percentage (10 %) - Frequency (10 out of every 100) - Framing - positive (90 % do not get the side effect), negative (10 % get the side effect), and both	Laypeople – web panel (437)	Between-subjects	- Format - Framing - Numeracy	Perceived risk of medication	- No overall difference in perceived risk based on number format. - Positive framing led to lower perceived risk than negative framing (no difference with both framings). - Participants with lower numeracy perceived percentages as less risky than frequencies.
Pighin et al. (2015) ⁴³ Italy	Interpretation of fetus DNA test	Single case probability in frequency format: - Distributive – “1 of 5 cases that shows signs of an anomaly and in which an anomaly is actually present” or “1 of the 2 cases that shows an anomaly, but no anomaly is present” - Absolute – “Probability fetus has an anomaly is 5 out of 7”	Laypeople – women subscribers to pediatric website (246)	Non-experimental	- Numeracy - Distributive or absolute probability	Understanding of fetal test results (restating numbers for distributive or calculating numbers for absolute)	- Participants correctly answered the distributive (frequency) question more often than the absolute (probability) one. - Higher numeracy was related to more accurate absolute understanding but not distributive.
Raghubir (2008) ²⁹ United States Only Studies 1 and 3 are included here.	Table showing frequencies of leading cause of deaths among 15–24 year olds	- Percentages - Varying denominators (e.g., 0.903/1000 or 90.3/100,000)	Study 1 (70) Study 3 (91)	Study 1: Between-subjects Study 3: Between- and within-subjects	Study 1: - Percentages, changing denominator Study 3: - Percentage with varying denominator	Study 1: - Perceived risk Study 3: - Perceived worry	- Study 1: Less perceived risk when framed with a larger denominator, making the percentage smaller. - Study 3: Perceived worry/risk was higher when the rate was higher (90.3/100,000) versus lower (0.903/1000).
Reen, Silber, and Langdon (2018) ⁴⁴ United Kingdom	Hypothetical treatments for disease	- Absolute terms (“100 more patients taking drug A will experience risk B”) - Relative terms (“2 times as many patients taking drug A will experience risk B”) - Numbers needed to treat/harm	Laypeople – from clinics diagnosed with MS (45)	Between-subjects, repeated measures	- Absolute terms - Relative terms - NNH - Numeracy	Understanding: answering correctly	- Understanding was greater for absolute terms compared with relative terms and NNH; no significant difference between relative terms and NNH. - Adding baseline information improved understanding. - Numeracy was related to better understanding.

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Table 3 (continued)

First Author (Year) Country	Study Context	Statistical Concept (s)	Respondents (N)	Design	Independent Variables	Primary Outcome Measures	Main Findings
Sevilla, Matthew, and Bagchi (2018) ⁴⁵ United States	Scenarios including: Sales performance, hypothetical drug, fictional library, mutual fund Experiment 5: Marketing cheese in store	(NNH) (“10 patients would have to take drug A for 1 patient to experience risk B”) • Baseline information (“150 patients taking drug A experienced risk B, 50 patients taking placebo experienced risk B”) • Percentage (e.g., “Product X is in the top 20 %”) • Frequency (e.g., “Product X is in the top 10”) • Set size (large > Top 100, small < Top 10)	Students Experiment 1 (233) Laypeople – web panel (Experiments 2–4) Experiment 2 (784) Experiment 3 (300) Experiment 2 (784) Experiment 4 (362) Experiment 5: Store customers (N/A)	Experiments 1–4: Between- subjects Experiment 5: Field experiment	Experiments 1–4: • Numerical format • Set size Experiment 5: Numerical format	Experiments 1–3: • Perception of product or library (favorable or not) Experiments 2 & 4: • Intentions to buy/ invest Experiment 5: • Quantity of cheese sold	• No differences for percentage or frequency. • Those in the small set size condition perceived the product/library more favorable with a frequency format than a percentage. The opposite was found in the large set size condition. • Experiment 5: More cheese was sold with percentage than frequency.
Siegrist and Keller (2011) ⁴⁶ Switzerland	Experiments 1 & 2: Hypothetical cancer diagnostics Experiment 3: Hypothetical social and cookie scenario Experiment 4: Hypothetical social and medical scenario	• Probabilities – single event (0.8 %) • Conditional probability (probability that woman with positive test result actually has cancer) • Natural frequencies (8 out of 1000) • Varying denominators	Laypeople – random from phone book Experiment 1 (266 females) Experiment 2 (148) Experiment 3 (206) Experiment 4 (257)	Between- subjects	All: • Frequency • Probability Experiments 1–3: • Education Experiment 4: • Scenario context • Frequency: Varying denominator	• Comprehension - answered questions correctly	• Participants answered more questions correctly for frequency than probability format though most could not answer correctly. • Probability led to overestimating risk and frequency led to underestimating risk. • Higher educated participants were more likely to answer the frequency problem correctly. • Participants were more likely to answer the same information presented in a social scenario correctly versus a medical one. • No difference in correct answers based on denominator.
Sirota and Juanchich (2019) ⁴⁷ United Kingdom	Experiment 1: Hypothetical probability of disease while traveling Experiments 2 & 3: Steroid side effect	Frequency • 1-in-X (e.g., 1 in 13) • N-in-X*N (e.g., 10 in 130)	Laypeople – web panel Experiments 1 & 2 (605) Experiment 3: (702)	All experiments: Between- subjects	All experiments: • Number format • Response order	All experiments: • Decision to cancel or maintain travel/ medication • Subjective probability of contracting disease/side effect (perceived risk)	• 1 in X format led to higher subjective probability for getting disease/side effect and deciding to cancel trip/ stop medication (Experiments 1 & 2). • In Experiment 3 only, lower education was associated with higher subjective probability for 1 in X than higher education.
Sirota, Juanchich and Bonnefon (2018) ⁴⁸	Hypothetical risk of a disease when traveling	Frequency • 1-in-X • N-in-X*N	Laypeople – web panel (975)	Randomly assigned to 1 of 5 parallel design experiments	• Format • Health literacy • Numeracy	• Subjective and numerical probability of contracting disease (perceived risk)	• 1-in-X format led to higher subjective probability to contracting disease, and

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Table 3 (continued)

First Author (Year) Country	Study Context	Statistical Concept (s)	Respondents (N)	Design	Independent Variables	Primary Outcome Measures	Main Findings
United Kingdom				and to one of the format conditions		Decision to cancel or maintain travel	decision to cancel trip, than N-in-X*N. More numerate less likely to overestimate; there was still a relationship with 1 in X format.
Skubisz (2019) ⁴⁹ United States	Hypothetical risk of violence	- Natural frequency (e.g., Of every 100 similar to student, 10 are estimated to commit act) - Simple frequency (e.g., Of every 10, 1 is estimated commit act) - Probability (e.g., Patients similar to student have a 0.10 probability of committing act) - Percentage (e.g., Patients similar to student are estimated to have a 10 % probability of committing act)	Students (414)	Between-subjects	- Natural frequency - Simple frequency - Probability - Percentage	- Perceived understanding - Risk judgments of harming - Judgments on how closely to monitor - Decision to expel - Feeling safe - Confusing Information	- Simple and natural frequency were easier to understand than probability. - Simple frequency rated safer than natural frequencies. - Probability was more likely to be evaluated as low risk than natural frequencies. - No difference between formats on monitoring or decisions to expel.
Sladakovic et al. (2016) ⁵⁰ Australia	Hypothetical medication	- Ranges/C.I. (e.g., 63–79 %) - Point estimates (e.g., 71 %)	Students (142)	Within-subjects	4 groups given harm/benefit information twice: First time: - Both point estimates - Harm - point estimate, benefit - range - Harm -range, benefit - point estimate - Both ranges Second time as opposite (e.g., Both ranges, etc.) - Numeracy	- Intention to take medication - Understanding – number of correct answers - Perceived harm/benefit - Severity	- Understanding was greater when information was conveyed as point estimates compared to ranges. - Those presented with point estimates after ranges had increased intentions to take medication. - Numeracy was associated with better understanding, higher intention to take medication, and lower perceived risk.
Thompson and Newman (2015) ⁵¹ United States	Hypothetical DNA/shoeprint evidence	- Frequencies/RMP (e.g., 1 in 100 people) - LRs (e.g., results are 100 times more likely to come from defendant than another) - Verbal equivalents (VE) (e.g., moderate support) - Magnitude or Strength	Laypeople – web panel (541)	Between-subjects	- Evidence (DNA or shoeprint) - Strength (very strong (e.g., 1 in 1 million), moderate (e.g., 1 in 100)) - Format (RMP, LR, VE)	- Understanding of data (questions correct) - Perceived understanding - Verdict rendered - Perceived guilt of defendant	- Format (RMP, LR, VE) had no main effects. - Strength/magnitude of the evidence related to conviction rate for RMP with DNA and shoe print evidence; but for LR it was only related to conviction rate for DNA evidence. - Perceived understanding was high for all conditions but most agreed with incorrect interpretations of the evidence
Woloshin and Schwartz (2011) ⁵² United States	Hypothetical drugs	- Frequencies (e.g., 20 in 1000) - Variable frequencies (VF) (e.g., 2 in 100; 2 in 1000) - Percentages (e.g., 0.02 %)	Laypeople – research panel (2,944)	Between-subjects	- Frequency - VF - Percent - Percent plus either types of frequency - Numeracy/education	Understanding -number answered correctly and pass/fail	- Understanding was lowest for frequency and VF and highest for percentage. - No differences for numeracy and education.

and intervals were presented as percentages⁵⁰ or as frequencies (20 out of 100 or 16–24 out of 100).³⁹ Confidence intervals also led participants to have a higher perceived risk and were rated less credible than point estimates and small ranges.³⁹ Han et al.³⁴ did not find a difference between confidence intervals and point estimates on perceived risk or credibility of information. They did find that participants were more worried about getting cancer when presented with a hypothetical confidence interval than a point estimate. This relationship held even when including comparative risk information.

3.1.4. Single or conditional event probability

Four studies examined an individual's ability to interpret or provide probability statements. Probabilities were generally more confusing and less easy to understand than frequencies^{43,46,49} and percentages.²⁸ Participants also rated probability statements as less serious²⁸ and conveying less risk⁴⁹ than frequencies and percentages. Participants were also more likely to view probability statements as an attempt to minimize the occurrence of events more so than percentages.²⁸ In contrast, Siegrist and Keller⁴⁶ found that participants shown conditional probability statements tended to over-rather than under-estimate risks. We note that the authors operationalized risk as a difference between the actual risk and the participant's calculated risk estimate rather than a perceived risk on a Likert-type scale as was used in other studies.

3.1.5. AR, RR, and NNT/NNH

Two of the three studies that examined AR, RR, and NNT/numbers needed to harm (NNH) found that participants could answer more questions correctly with an absolute risk format than relative risk and NNT/NNH formats.^{35,44} In one qualitative study, participants preferred the ARR format as it was easier to make comparisons. The RRR format was rated second, outperforming NNT/NNH or odds ratios.³⁵ In contrast, LaVallie et al.³⁷ found that participants were able to answer more questions correctly with a RRR format. We note that these studies used percentages for relative risks, but two used frequencies for absolute risks^{37,44} while the other used percentages.³⁵

3.1.6. Likelihood ratios

Two articles examined participants' comprehension of likelihood ratios using forensic evidence (e.g., 100 times more likely to come from a suspect than another random person).^{30,51} Both studies compared likelihood ratios to random match probability (a form of frequency such as "1 in 1 million"). Thompson and Newman⁵¹ found that while participants believed they understood the meaning of both likelihood ratios and random match probabilities, more than half agreed with incorrect interpretations of the evidence. In contrast, Bali et al.³⁰ found that participants were less likely to correctly answer questions in the likelihood ratio format than the frequency format.

3.1.7. Odds ratios

Only one study (qualitative) examined the interpretation of odds ratios and found that participants struggled with interpreting odds ratios. Over half of the participants considered odds ratio statements unclear.³⁵ Further, participants did not prefer this format compared with ARR, RRR, or NNT.

3.2. Cross-cutting concepts

In addition to examining statistical concepts, some of the articles explored other factors that may impact comprehension. This section summarizes these types of studies examining magnitudes, framing of impact (benefits vs. harm), and numeracy.

3.2.1. Varying magnitudes

Studies looking at magnitude examined the size or value of various statistical information usually ranked from small to large.^{28,32,36,41,51}

Kupor and Laurin³⁶ found that when comparing low and high

percentages of probable outcomes of a product (e.g., "7 % of users will experience a cough" vs. "68 % of users will experience a cough"), participants erroneously perceived that the higher percentages meant they would experience those outcomes at a greater magnitude (i.e., more frequent coughing within an individual).

Three articles found that magnitude had a moderating effect on interpretation. Oudhoff and Timmermans⁴¹ found that when using small magnitudes (e.g., 0.67 %) there was no difference between frequencies or percentages on participants' perceptions of the likelihood of winning the lottery. However, when using large magnitudes (e.g., 25 %), the frequency with the smallest denominator (e.g., 1 in 4) was viewed to have a higher likelihood than percentages or a variable denominator format (e.g., 25 in 100). Brase²⁸ found that for large magnitudes, frequencies and percentages were deemed more serious than equivalent point estimates (e.g., 90 million of all Americans) or probabilities (e.g., probability of 0.33), but the opposite was true for small magnitudes, where point estimates were considered to be most serious (e.g., 2.7 million of all Americans versus probability of 0.01). In the third study, when forensic evidence was strong or had higher magnitudes, participants were more likely to convict when presented with frequencies for both DNA and shoeprint evidence, but when presented with likelihood ratios, participants were only more likely to convict with the DNA evidence.⁵¹

DeWolf et al.³² examined how individuals compared different number formats. They found that participants had a hard time comparing the magnitude of values (i.e., higher error rates and longer response times) when the values were in a fraction format but performed better when the values were in integer or decimal format.

3.2.2. Framing

A few articles explored how framing the statistical information as a risk/cost or benefit would impact perceptions. In a qualitative study, Ilic et al.³⁵ found that participants preferred positively framing the information in terms of surviving versus dying. Peters et al.⁴² found that when providing positively framed information, participants perceived the medication as less risky than with a negative frame. Mohan et al.⁴⁰ explored how participants perceived percentage changes when they were framed as a cost or benefit. They found that even when the percent reduction or cost was a superior situation, participants did not perceive that option as better. For example, their study suggested that it is better to frame a sale as a 25 % bonus product rather than a 25 % discount, as consumers tended to pick the 25 % bonus benefit option even though it would not be a better deal for them.

3.2.3. Numeracy

Approximately half of the articles examined whether participants' numeracy, or the ability to understand and use math and numerical concepts, had an impact on their understanding of statistical concepts. Most of these articles found that individuals with higher numeracy were better able to comprehend statistical concepts such as likelihood ratio, frequencies, ARR, RRR, NNT, and confidence intervals.^{30,37,38,40,43,44,50} Pighin et al.⁴³ found that while higher numeracy was related to more accurate comprehension of probabilities, it did not increase the accuracy of understanding frequencies.

Some studies also found that higher numeracy is related to lower perceived risks⁵⁰ or estimated risks,⁴⁸ and that it equalized the differences in perceived risks between percentages and frequencies.⁴² Three articles found no impact of numeracy on comprehension or perceived risk. Each noted that their sample was either high on numeracy or relatively low on numeracy, and thus, the absence of a relationship could be due to homogeneous samples.^{34,41,52} Finally, numeracy was also associated with increased intentions to take a medication.⁵⁰

4. Discussion and conclusion

4.1. Discussion

This literature review identified themes that support evidence from previous literature reviews as well as identified findings that differed from previous literature review findings. Overall, the evidence from these articles suggests that some statistical concepts are consistently misunderstood. Likelihood ratios, odds ratios, NNT/H, and confidence intervals tend to cloud laypeople's understanding of quantitative information. Similar results were found in other literature reviews.^{10,12,53}

In this review, probabilities were also consistently less well understood; however, we classified articles as examining probabilities only if these articles used the word “probability” in the scenario to participants such as, “A drug about to be approved by the FDA has been estimated as having a probability of 0.99 for being effective for any given American.”²⁸ Some studies used other phrasings and synonyms to “probability,” such as “risk of” or “chances of.” For example, Damman et al.³¹ used “Your risk of cardiovascular disease is 27 %” while Han et al.³⁴ noted, “Your chances of developing colon cancer in your lifetime are 9 %.” It is possible that merely using the word “probability” rather than “chances of” or “risks of” decreases understanding for laypeople, as percentages were better understood. Further, Skubisz⁴⁹ used the word “probability” in their percentage condition and found there was no difference in understanding between percentages and probabilities. The potential for these wording differences to impact comprehension warrants further research.

The results are mixed for whether conveying risks as frequencies or percentages is better for comprehension. This finding differs slightly from Akl et al.¹⁰ who found that frequencies were better understood than percentages. Our results did confirm past reviews showing that the use of consistent denominators improves understanding.^{11,14} Further, if frequencies are used, the scaling of the denominator impacts comprehension of results, such that a small denominator is perceived as having a much bigger impact (e.g., greater risks or more benefits) than a large denominator. These results suggest that if framing information in terms of benefits, a small denominator would be perceived as a greater benefit than if using a large denominator, especially if using frequencies. The opposite would be true if framing information in terms of risks. In addition, it remains unclear if there is a threshold where frequencies or percentages hold an advantage. Sevilla et al.⁴⁵ found 100 may be the threshold; that when the denominators are small or less than 100, frequencies were more impressive but when they were more than 100 (e.g., 20 out of 200); percentages were more impressive. Future research may explore if this finding is sustained with risk information and investigate not only how individuals interpret risks versus benefits in these formats but also explore how this interpretation may impact decision-making.

Results show that ARR or RRR are better understood than NNT, but they are mixed for what is better between ARR and RRR. The studies showed a slight preference for ARR by participants. This could be due to how these studies operationalized absolute and relative risks. Although all three studies used percentages for providing relative risks, two used frequencies for the absolute risks^{37,44} while the other used percentages.³⁵ These results align with previous studies that have shown that both ARR and RRR are better understood than NNT.^{10–12,53} Prior literature reviews also found inconsistency in preference between ARR and RRR; with a slight case for RRR being more persuasive¹⁰ and perceived more favorably⁵³ but with ARR related to better understanding.¹² Covey⁵³ also noted that these differences could be due to how the scenarios were worded.

The differences across studies and findings could be due to a series of qualifying factors such as numeracy, framing, and operationalization of the outcome measures.

4.1.1. Numeracy

Individuals with low numeracy have an increased susceptibility to

the format of how information is presented and to subsequent biases in interpretation. Even with numeracy being measured in many different ways, the results were relatively consistent across articles. Because it may be difficult to tailor DTC promotional materials to people with high versus low numeracy, it is important to determine the best way to present the information for all consumers.

4.1.2. Framing and magnitude

How risk is framed influences laypeople's perceptions of risk and decision-making. It can be framed positively, such as the probability of not experiencing a side effect or the estimate of surviving. Or it can be framed negatively, such as the probability of experiencing a side effect or dying. The evidence points to laypeople preferring positively worded risks and perceiving the risk to be lower than when the same information is presented negatively. The type of scenario may also matter. Siegrist and Keller⁴⁶ found social scenarios were better understood than medical scenarios when the same kinds of statistics were presented. Therefore, it may be even more important to investigate how to communicate medical risk information in a way laypeople understand. The articles also point out how the magnitude of the risk estimates can impact interpretation, with laypeople being confused on how to interpret different levels of, for example, confidence in risk statements.

4.1.3. Operationalization of outcome measures

Across the articles, comprehension and other interpretation measures were assessed in different ways, making comparisons more difficult. “Understanding” or “comprehension,” for example, was operationalized as self-rated understanding or clarity^{28,35}; the number of questions answered or interpreted correctly^{36,37,39,44}; the ability to recall the information³¹; or the ability to calculate or determine numbers based on the data presented.^{43,46} It is clear that there is no agreed-upon measure for assessing comprehension and that it varies based on the statistic and scenario. Bali et al.³⁰ proposed a set of comprehension measures coined the “CASOC framework,” which ranged from assessing if a participant was able to give equal weight to evidence of equal strength to whether participants updated their beliefs based on new evidence. Further, it was not always clear if recall was a factor when assessing participant comprehension. For example, were participants able to see the statistical information while completing the comprehension questions, or was it on the next page or computer screen? Finally, the articles examined other outcomes using various measures for determining perceived risk^{29,31,34,42} and behavioral intentions.^{28,41,45,48} These different measures may have impacted the findings of the studies.

Ancker et al.⁵⁴ have also found it difficult to draw conclusions about the impact of the different statistical concepts given the various ways to operationalize outcomes such as understanding or comprehension. The authors introduced a taxonomy to help understand the conflicting results from these types of studies including outcome measures, data formats, and data structures. The outcome measures taxonomy, more relevant to this section, lists the different ways to assess the impact of including statistical concepts including sensory processing, perception (i.e., affective, perceived magnitude, cognitive perceptions, and perceptions of communication), memory or recall, decision or behavior intention, and action (actual behavior). Akin to what was found in this review, Ancker et al.⁵⁴ proposed that researchers should use similar nomenclature when assessing different outcomes rather than just using terms such as comprehension or understanding. However, there are not enough studies in each category of their taxonomy in our review to make conclusions based on these categories. It is possible that future research will enlist the aforementioned taxonomy for outcome measures to assess the various aspects of comprehension. Similarly, future research should be clear about whether assessments of comprehension also assess participants' recall.

4.1.4. Research gaps for future studies

We found no studies that examined the interpretation of common statistical concepts such as minimum or maximum, median, mean, mode, power, or statistical significance. Second, we identified no studies on less common terms such as hazard ratio. It would be interesting to determine if common terms (e.g., average, mean, statistical significance) that may be assumed to be understood by consumers are understood as intended. Third, it would be useful to determine if consumers understand clinical trial concepts that often accompany statistical terms in medicine, such as “control group.” In addition, despite existing reviews, researchers do not always agree on the best use of the statistical measures discussed in this review article. For example, some researchers still advocate for odds ratios,⁵⁵ while others are proponents for relative risk (RR).⁵⁶ Comprehensive studies are needed to address these additional terms and continue to test the statistical concepts in this review (e.g., frequencies, odds ratios, likelihood ratios). Further future studies should strive to use a consistent operationalization of the statistical concepts and of comprehension, which may include defining comprehension in a multidimensional way. Finally, DTC promotional claims often combine statistical terms with each other and with qualitative terms like “some” or “at least.” Future research could explore consumers’ interpretation of these complex real-world claims.

4.1.5. Literature search limitations

There are some limitations for this literature review. First, we did not include any article that was not in the English language; however, we did not limit the search by country. Second, we did not include any studies that relayed statistics visually, despite the fact that research has shown that graphs help with interpretation of statistics.^{12,14} Our focus was on applying these results to DTC promotion, which may not have the space for graphic representations. Last, we did not assess the quality of the studies conducted nor the potential bias of the studies, as this literature review focused on describing the available research related to lay-audience comprehension of statistical concepts. Recognizing these limitations, we contend that the studies identified still provide insight on the current state of research in this field.

5. Conclusion

The findings from our literature review show that there is still much to learn about how lay audiences interpret statistical concepts, particularly those that could appear in DTC promotion. We also identified gaps in the literature, finding an absence of any articles on statistical concepts such as “mean,” “median,” or “statistical significance.” Future research could explore these additional statistical concepts and how quantitative outcomes are framed and operationalized. Additional research could help determine the best methods for presenting understandable information to laypeople so they can make educated decisions. To help improve equity in making health decisions, research could focus on ways to improve the presentation of statistical and quantitative information to individuals with low numeracy.

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Declaration of competing interest

None

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Appendix A. Supplementary data

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